FLEETWISE NEXUS

Autonomous Fleet Intelligence & Dispatch Orchestration System

Comprehensive Technical Documentation // System Architecture & Verification Manual

Stack: FastAPI, SQLite WAL, React 19, TypeScript	**Isolation:** BEGIN IMMEDIATE (Row-Lock)	**Security:** SHA-256 Chain + RSA-4096	**Status:** Production Ready (v2.0)

1. Executive Summary & Core Pillars

FLEETWISE NEXUS is an enterprise-grade autonomous fleet management and dispatch orchestration platform. Engineered with a high-throughput **FastAPI** backend, an atomic **SQLite WAL (Write-Ahead Logging)** persistence engine, and a modern **React 19 + TypeScript + Tailwind CSS** responsive frontend, the system guarantees 100% deterministic dispatching, cryptographic compliance verification, and autonomous incident recovery under high concurrency.

Pillar	Architectural Guarantee	Key Mechanism
Zero-Collision Concurrency	Zero double-booking of vehicles or operators under simultaneous threads.	SQLite `BEGIN IMMEDIATE` transactional locks
Safety Passport Audit	8-vector pre-dispatch compliance verification before any mission launch.	Mathematical vector evaluation + SHA-256 seal + RSA-4096 signature
Explainable Smart Match	5-factor AI matching optimizing distance, load, safety, and carbon.	30% Proximity, 20% Capacity, 20% Safety, 15% Carbon, 15% SLA
Autonomous Failover	Zero-downtime mission recovery upon breakdown or operational anomaly.	Autonomous custody chain transfer & emergency reserve dispatch
Dispatch Black Box	Cryptographically chained tamper-evident flight-recorder audit trail.	SHA-256 blockchain hash link verification across all dispatch events

2. System Architecture & Concurrency Model

The system employs a strict 3-tier decoupled architecture designed for deterministic latency, zero race conditions, and seamless UI state synchronization:

Layer	Technologies	Responsibilities & Guarantees
Client Tier	React 19, TypeScript, Vite, Tailwind CSS v4	Single-page responsive dashboard; zero-overlap cards; debounced scrubber (150ms); 20s quiet background polling.
Application Tier	FastAPI, Uvicorn, Pydantic v2, Python 3	Asynchronous REST API; strict input sanitization; 8-vector audit engine; multi-threaded attack lab; auto-recovery orchestrator.
Database Tier	SQLite 3 with WAL Mode & Foreign Keys	ACID serializable persistence; PRAGMA busy_timeout=5000; non-blocking readers; atomic row-level locks via BEGIN IMMEDIATE.

Database WAL PRAGMA Configuration:

```
PRAGMA journal_mode=WAL; PRAGMA busy_timeout=5000; PRAGMA foreign_keys=ON; PRAGMA synchronous=NORMAL;
```

This configuration ensures writers never block readers, reads proceed concurrently without locks, and write transactions automatically wait up to 5,000ms before encountering busy conditions.

3. Core Signature Features & Algorithms

Signature Feature 1: The 8-Vector Assignment Safety Passport

Prior to authorizing any dispatch mission, the engine calculates a composite audit score across 8 distinct operational vectors. If any mandatory vector fails (such as missing driver endorsements or inadequate vehicle refrigeration), the assignment is immediately aborted.

Vector Code	Evaluation Dimension	Wt	Pass / Fail Criteria
VEC-1-FATIGUE	Driver Hours of Service & Fatigue Margin	15%	PASS if logged hours + trip < max (10h) and fatigue < 75. CONDITIONAL if > 60.
VEC-2-CERT	Cargo Certification Endorsements	20%	MANDATORY PASS: Driver must hold 100% of required endorsements (Cold Chain, Hazmat).
VEC-3-CAPACITY	Payload Weight & Volume Tolerances	15%	PASS if payload weight ≤ vehicle capacity and volume ≤ cargo volume. CONDITIONAL if > 85%.
VEC-4-RANGE	Energy Reserve Buffer (125% Rule)	15%	PASS if range ≥ distance × 1.25. CONDITIONAL if reserve margin < 25%. FAIL if range < dist.
VEC-5-MAINT	Mechanical Integrity & Inspection Score	10%	PASS if inspection rating ≥ 75/100 and odometer within maintenance due window.
VEC-6-SAFETY	Operator Historical Telematics Index	10%	PASS if telematics safety score ≥ 85/100 (clean acceleration, smooth braking records).
VEC-7-SECURITY	Cargo Equipment & Environmental Match	10%	MANDATORY PASS: Cold Chain requires verified onboard Cryo unit; High Value checks armored specs.
VEC-8-HAZARD	Dynamic Route Hazard & Atmospheric Clearance	5%	Real-time corridor clearance checking pavement friction, zero wind shear, visibility > 10km.

Signature Feature 2: Explainable Smart Match AI Heuristic

Smart Match evaluates all active fleet combinations using a calibrated 5-factor heuristic formula:
Composite Score = 0.30 × Proximity + 0.20 × Capacity + 0.20 × Safety + 0.15 × Carbon + 0.15 × SLA
Candidates lacking mandatory cargo hardware (e.g. Cryo unit for biologicals) are filtered before scoring. The API generates plain-English rationale bullets and transparency tradeoff disclosures explaining the selection.

Signature Feature 3: Double-Booking Attack Lab (Concurrency Defense)

To rigorously prove serializability, the Attack Lab launches 15 to 30 concurrent threads targeting the exact same vehicle and driver at the identical millisecond. By wrapping the assignment in a transactional **BEGIN IMMEDIATE** block, SQLite locks the target rows. **Exactly 1 dispatch receives HTTP 201 Created**, while all remaining requests receive deterministic **HTTP 409 Conflict** exceptions. The database state remains completely intact with 0 double-bookings.

Signature Feature 4: Autonomous Recovery Dispatch

Upon simulated operational failure (e.g., Mechanical Failure, Driver Fatigue, Route Obstruction), the Auto-Recovery protocol executes: 1) Flags the active mission as Recovered; 2) Locks the disabled vehicle for Maintenance; 3) Executes an emergency Smart Match query for nearest reserve assets; 4) Issues a new failover assignment (ASN-REC-XXXXXX) with an updated Safety Passport and continuous Black Box audit trail.

4. Temporal Simulation, Telemetry & Black Box Ledger

Feature	Functionality	Performance & Architectural Detail
Fleet Time Machine	Interactive -24h to +24h historical and future scrubber.	Linear battery discharge modeling; bottleneck predictions; debounced 150ms fetch prevents network flooding.
Availability Heatmap	168-Hour (7 Days × 24 Hours) capacity strain matrix.	Calculates fleet capacity strain across time slots: Idle (<40%), Normal (40-65%), Optimal (65-85%), Peak (>85%).
Dispatch Black Box	Cryptographically chained SHA-256 flight-recorder ledger.	Each block links to the previous block's SHA-256 hash. Real-time /api/black-box/verify-chain confirms ledger integrity.
Fleet Pulse	Real-time live fleet telematics and operational KPI strip.	Live readouts for Active Fleet Utilization %, CO2e Avoided kg, SLA Reliability %, and Driver Readiness.
Cargo Commissioning	Dynamic consignment portal with automatic endorsement pairing.	Full CRUD persistence to SQLite; supports Cold Chain, Hazmat, High Value Escort, and Heavy Cargo presets.

5. Complete REST API Endpoint Specification

Method	Endpoint Route	Code	Operation & Description
GET	/api/health	200	Engine operational status and version
GET	/api/fleet-pulse	200	Real-time telemetry, KPIs, and alerts
GET	/api/vehicles	200	List vehicles with optional status filter
POST	/api/vehicles	201	Commission and register a new vehicle asset
GET	/api/drivers	200	List certified operators and safety scores
GET	/api/deliveries	200	List active and pending cargo consignments
POST	/api/deliveries	201	Commission new cargo consignment (persisted to DB)
GET	/api/assignments	200	List active dispatch missions and passports
POST	/api/assignments	201/422	Dispatch mission with 8-vector Safety Passport audit
POST	/api/assignments/{id}/complete	200	Complete mission and release vehicle/driver to Available
GET	/api/safety-passports/{asn_id}	200	Retrieve official signed 8-vector Safety Passport
GET	/api/smart-match/{delivery_id}	200	Calculate multi-factor explainable AI match recommendations
POST	/api/attack-lab/run	200	Execute multi-threaded concurrency attack stress probe
POST	/api/recovery/trigger	200	Trigger automated incident failover protocol
GET	/api/time-machine/snapshot	200	Simulate fleet state across -24h to +24h offset
GET	/api/heatmap	200	Retrieve 168-hour capacity strain matrix
GET	/api/black-box	200	Query cryptographically linked immutable flight-recorder log
GET	/api/black-box/verify-chain	200	Verify linear integrity of SHA-256 blockchain hash chain
POST	/api/reset-demo	200	Reinitialize pristine demo seed dataset

6. Automated Testing & Verification Suite

The platform includes a 100% automated test suite executed via Pytest (`PYTHONPATH=. pytest backend/tests/`). All tests pass deterministically in under 0.30 seconds:

Test Case	Target Functionality	Observed Result
test_health_check	API liveness and engine status verification	PASSED — HTTP 200 OPERATIONAL (v2.0.0)

test_list_vehicles / drivers / deliveries	Entity registry queries & Pydantic schema validation	PASSED — Pre-seeded luxury fleet retrieved
test_smart_match_scoring	5-factor AI heuristic composite score calculation	PASSED — Ranked candidates returned with rationale
test_safety_passport_audit_and_dispatch	8-vector pre-dispatch audit & RSA/SHA signature	PASSED — Issued cryptographic passport hash seal
test_black_box_hash_chain_verification	SHA-256 blockchain tamper-evident validation	PASSED — Chain verified from genesis to head
test_auto_recovery_incident	Emergency failover upon vehicle breakdown	PASSED — Failover mission created & assets locked
test_atomic_concurrency_attack	20 concurrent threads attacking identical vehicle/driver	PASSED — Exactly 1 dispatch succeeded, 19 blocked

7. Local Setup & Quickstart Guide

1. Clone Repository:

```
git clone https://github.com/Sowparnika786/fleet.git && cd fleet
```

2. Start Backend API Daemon:

```
python3 -m uvicorn backend.main:app --host 0.0.0.0 --port 8000 --reload
```

3. Start Frontend Development Server:

```
npm install && npm run dev -- --host 0.0.0.0 --port 5173
```

4. Access System:

• Web Dashboard: **http://localhost:5174/** (or http://localhost:5173/)
• OpenAPI Swagger Documentation: **http://localhost:8000/docs**
• Pytest Suite: **PYTHONPATH=. pytest backend/tests/**